Final Course Project Guidelines

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Worth **20%** of the course grade

META-PROJECT THEME

Tiny Deep Learning Models — doing more with fewer resources

Every project sits under this theme. You choose your own topic within it, and **the topic must be approved by the instructor before you begin**. Think constrained compute, small datasets, model compression, quantisation, distillation, efficient architectures, or on-device inference.

01 Team Formation

- **3–4 students** per group.
- **Equal contribution from every member is mandatory** and must be documented.
- Each member states their **individual contributions** in a named section of the final report.

Note. A report without a per-member contribution statement is treated as incomplete. Contribution records may be checked against the repository's commit history.

02 Choose Your Track

Pick one goal at team-selection time. It decides which written deliverable you produce.

TRACK A · DEFAULT

Technical Report

- Code repository
- Technical report, max 10 pp.
- Oral presentation
- Research poster

TRACK B · CONFERENCE

Research Paper

- Code repository
- **Research** report — not a technical one
- Oral presentation
- Research poster

TRACK C · HACKATHON

Poster & Pitch

- Code repository
- Research poster
- **Pitch** presentation

03 Written Report

FORMAT

ACL submission template, via Overleaf:
overleaf.com/read/mknhysghvcgh

REQUIRED SECTIONS

- **Abstract** — 150–250 words
- **Introduction & Motivation**
- **Related Work**
- **Data Description & Preprocessing**
- **Methodology**
- **Experimental Results & Discussion**
- **Conclusion & Future Work**
- **References**

Track B teams: write this as a **research paper** — a contribution, a hypothesis, and experiments that test it — not as a project write-up.

04 Starting Points

Directions that fit the theme. These are prompts, not a menu — bring your own topic and have it approved.

- **Compression** — pruning, quantisation or low-rank factorisation of a trained model, measuring the accuracy/size trade-off.
- **Knowledge distillation** — training a small student network from a larger teacher.
- **Efficient architectures** — depthwise-separable convolutions, MobileNet- or TinyML-style designs.
- **Low-data regimes** — transfer learning, augmentation, or few-shot methods where labelled data is scarce.
- **On-device inference** — deploying to a phone, Raspberry Pi or microcontroller, with latency and memory reported.

Report the cost, not just the accuracy. Under this theme a result is only meaningful alongside its parameter count, model size, and inference latency.

05 Presentation & Poster

ORAL PRESENTATION

- **12–15 minutes** plus Q&A.
- **All members must present.**
- Professional slide design; a **live demo is encouraged**.

RESEARCH POSTER

- **A1 or A0** format.
- Visual representation of the work, with key findings highlighted.
- **QR code** linking to the repository.

06 Evaluation Rubric

CRITERION	WHAT EARNS FULL MARKS	WEIGHT
Technical Implementation	Correct, complete, well-organised implementation	**20%**
Model Design & Performance	Well-justified architecture; strong performance on appropriate metrics	**20%**
Innovation & Relevance	Original, well-motivated and highly relevant solution	**20%**
Experimental Analysis	Thorough analysis, comparisons and critical interpretation	**20%**
Presentation & Report	Clear, professional and technically accurate	**20%**
Total		**100%**

Each criterion is graded Excellent / Good / Fair / Poor. The full descriptor table is posted on Blackboard.

07 Submission

FILE NAMING

```
CS4085_Project_GroupX_TopicY.zip
```

REQUIRED CONTENTS

- Technical report — **PDF, ACL format**
- Presentation slides — **PDF or PPTX**
- Poster — **PDF, A1 or A0**
- GitHub repository link — **in the README**
- Portfolio website URL — **in the README**

Deadline policy. Missing the submission deadline results in losing full marks for this assessment. **AI-generated content must not exceed 25%** of any submitted work, and must be properly cited.

08 Milestones

WEEK	MILESTONE
Week 2	Team formation and **topic approval**
Week 5	**Project Part 1 due**
Week 6	In-class project work
Week 15	**Final presentations** and full submission

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